DETERMINANTS OF MALNUTRITION AMONG UNDER-FIVE CHILDREN IN KENYA

APPLICATION OF THE MODIFIED POISSON REGRESSION APPROACH

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INTRODUCTION

- Malnutrition is defined as imbalance, deficiencies, or excess nutrients intake. It is classified as undernutrition or overnutrition
- Undernutrition is either stunting (low height for age), underweight (low weight for age), and wasting (low weight for height) (Kenya: Nutrition Profile, 2021).
- Proper nutrition is a crucial factor for the growth a development of children under five.

PROBLEM STATEMENT

- In Kenya, malnutrition rates are a cause for concern, with high numbers in certain regions. It exposes children to various infections, which complicate the outcome
- Results from the Kenya Demographic Health Survey (2014/2015) indicate that approximately 26% of the children under five are stunted, 4% are wasted, and 11% are underweight.

OBJECTIVES

General Objective

To determine the factors associated with under-five malnutrition in Kenya using the KDHS 2014 dataset.

Specific Objectives

- To estimate the prevalence of malnutrition among children under-five in Kenya.
- To determine the association between bio-demographic and socio-economic factors and under-five malnutrition.
- To fit a multivariate model to determine factors associated with under-five malnutrition.

JUSTIFICATION

This study was aimed at providing awareness of the factors associated with malnutrition among children under five.

LITERATURE REVIEW

- Several policies and measures have been put in place in the country to address the issue of malnutrition. Although these policies have helped counter the rising rate of malnutrition, cases are still persistent.
- Socioeconomic factors associate social and economic factors. They are lifestyle elements and measurements of both social standing and financial viability. These include: maternal nutrition, maternal education, wealth index and place of residence.
- Bio-demographic factors are factors in which biological considerations such as evolutionary, epidemiological and genetic determinants are accentuated. Child health and survival can be affected by alterations in reproductive patterns through various mechanisms, more so through changes in; birth order, maternal age at birth or birth intervals (Sathiya Susuman et al., 2016).

METHODOLOGY

- The study used the Kenya Demographic Health Survey (KDHS 2014) dataset. Specifically, the interest was with the Children's recode (KEKR72FL).
- Descriptive statistics were done to summarize the data quantitatively.
- The summary statistics also gave a general outlook of the data.
- Bivariate analysis was done to examine the strengths of the relationship among variables of interest.
- For multivariate analysis, the model applied was the Modified Poisson Regression model.

RESULTS

- From the results, 26.01% were stunted and 10.87% were underweight.
- A majority of stunted children (15.08%) and underweight children (7.2%) were from low income households (lowest and second lowest categories).
- Rural areas had a higher proportion of malnourished children than urban areas where 19.2% of children were Stunted and 8.48% were underweight.
- Males had a higher proportion of stunting (15.06%) and underweight (6.08%) as compared to females who had 10.95% stunted and 4.79% underweight.

Variable	Stunting IRR	p>Z	Underweight IRR
Maternal Education			
No education	1.223095	0.203	1.998971
Primary	1.583728	0.002	1.741453
Secondary	1.188813	0.258	1.156105
Type of Place of Residence			
Rural	0.8772408	0.027	0.8329995
Wealth Index			
Poorest	2.246326	0.000	3.231934
Poorer	1.779475	0.000	2.11903
Middle	1.563191	0.000	1.855637
Richer	1.50243	0.001	1.589544
Size of Child at Birth			
Very large	0.7957832	0.307	0.4255728
Larger than average	0.8597511	0.411	0.5179529
Average	0.9571711	0.807	0.679164
Smaller than average	1.360186	0.095	1.261039
Very small	1.610869	0.015	1.503941
Sex of the Child			
Male	1.366043	0.000	1.362328

RESULTS cont'd

- Mothers who attained primary level of education were 1.5 times more likely to have stunted children and 1.7 times more likely to have underweight children as compared to those with a higher level of education.
- Children born and living in rural areas are 1.88 times more likely to be stunted and 1.83 times more likely to be underweight as compared to children born and living in urban areas.
- Children from the poorest households were 2.2 times more likely to be stunted and 3.23 times more likely to be underweight than children from the richest households.
- Children who were very small at birth were 1.6 times more likely to be stunted than those who were larger in size at birth.
- Maternal age was also a significant predictor of malnutrition. Mother's aged 15-19 were 2.16 times more likely to give birth to stunted and underweight children as compared to mothers aged 45-49.

DISCUSSION

- In the multivariate Modified Poisson model, wealth index, type of place of residence, sex of the child, size of the child at birth, maternal education were significantly associated with both Stunting and Underweight among children under five.
- According to a study by Ambel in 2014, education determines the amount of health knowledge a mother has which is equally essential in child health.
- In urban areas, maternal health services such as ANC, PNC visits and health facility delivery are readily available and can equally raise community awareness and provide child immunization services and quality complementary feeding (Bado & Susuman, 2016) .
- Low income hoseholds have limited resources and thus less likely to access health facilities, expand their knowledge on access to health services and better nutritional practices that ensure children get a balanced diet (Bhagowalia et al., 2012).

CONCLUSION

- From the study, it was notable that the prevalence rates of malnutrition among under-fives were still a cause for concern and need to be addressed.
- Also, socio-economic and bio-demographic characteristics such as type of place of residence, region, wealth-index , birth-order, maternal age, size of the child at birth and maternal education were found to be significant in determining a child's nutrition status.
- The study revealed that low income households, residing in the rural areas, with primary education level and lower contribute the highest cases of under-five malnutrition in Kenya.
- Addressing these factors will likely reduce the cases of malnutrition.
- The factors that influence malnutrition cases in Kenya are interrelated and thus there is no overall blanket program that can solve the problem at once.
- A more rigorous approach is looking at the problem from a multi-disciplinary perspective.

RECOMMENDATIONS

- A greater majority of the bio-demographic predictors that are maternal related are more personal. The most effective government intervention is civic education. The Ministry of Health in conjunction with county governments should carry out campaigns , general advocacy and advertisements .
- The region differences can be addressed by increasing the funding dedicated towards healthcare.
- There is need for further research that uses primary data where variables that were not captured are incorporated.

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